

The empirical recurrence for OEIS A299718, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A299718 records “Empirical recurrence of order 83” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 130$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{83} = 33169152 \neq 0$, so no further tail satisfies anything shorter; any order-83 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299718 is “Number of nX5 0..1 arrays with every element equal to 0, 2, 3, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 127, 1205, 23877, 439647, 8668148, 174287007, 3523382014, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Feb 17 2018 10:39:56, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 130$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 130$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 83: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 83.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 260$ exact terms from $n = 4$ on, it returns order 83 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	17	c_{22}	588554862887	c_{43}	−78875399089484	c_{64}	93700568112300
c_2	168	c_{23}	−6991924616	c_{44}	−381049928114263	c_{65}	90485057614790
c_3	−1291	c_{24}	−1351256151658	c_{45}	605502188458579	c_{66}	−3121925931111
c_4	−16592	c_{25}	1814985126407	c_{46}	−408580547370284	c_{67}	−25868453828006
c_5	10953	c_{26}	−646264606356	c_{47}	−1284380121488537	c_{68}	−21671772474629
c_6	520302	c_{27}	−11315155080042	c_{48}	2063424466217903	c_{69}	−4216469792778
c_7	1014628	c_{28}	5741376781370	c_{49}	1499540196586662	c_{70}	9021260843523
c_8	−6244466	c_{29}	32447916657292	c_{50}	−4033079595570501	c_{71}	4336355986181
c_9	−23239972	c_{30}	−17264615346861	c_{51}	−837693298673134	c_{72}	−1265770714786
c_{10}	28604091	c_{31}	−28582306533994	c_{52}	4542597981118832	c_{73}	−1089256936768
c_{11}	208122811	c_{32}	81086763241525	c_{53}	−265543538877551	c_{74}	−185562993952
c_{12}	−42561451	c_{33}	−22958033404498	c_{54}	−2735267463499324	c_{75}	40455421000
c_{13}	−1120618121	c_{34}	−81985148704203	c_{55}	936349655928806	c_{76}	44842280130
c_{14}	454494822	c_{35}	110793936230317	c_{56}	643816708206647	c_{77}	−13905243312
c_{15}	5795809445	c_{36}	−287854437524078	c_{57}	−923756880473482	c_{78}	−4377875896
c_{16}	−4958338305	c_{37}	−230607133350082	c_{58}	129675327543114	c_{79}	8848124400
c_{17}	−32371309770	c_{38}	769771501533195	c_{59}	475621130348278	c_{80}	1154978144
c_{18}	24418675641	c_{39}	225551160758648	c_{60}	−26326995774076	c_{81}	−989461024
c_{19}	122782684359	c_{40}	−733745066544681	c_{61}	−53474991506721	c_{82}	−68424768
c_{20}	−119039414401	c_{41}	−104775271627987	c_{62}	−131193723235958	c_{83}	33169152
c_{21}	−221583273621	c_{42}	526236390400244	c_{63}	−107397131349539		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A299718 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{83} - c_1 x^{82} - \dots - c_{83}$. Its constant term is $-c_{83} = -33169152$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 83 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 83, so $\deg q = 83$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 83, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 83, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 33169152.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-83 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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